

A/B Test Analysis Report

Landing Page Optimization Experiment

Conversion Rate Analysis | Statistical Testing | Business Impact Assessment

Project	Project 3 — A/B Test Analysis & Experimentation
Tools Used	Python, MySQL, Power BI
Dataset Period	January 2017 (4-Week Experiment)
Significance Level	$\alpha = 0.05$ (Two-tailed Z-Test)
Verdict	DO NOT SHIP — New landing page does not improve conversions

1. Executive Summary

This report presents the complete analysis of an A/B test conducted to evaluate whether a new landing page design improves user conversion rates compared to the existing control page. The experiment ran over a 4-week period and involved 290,584 unique users split nearly equally between the control (old page) and treatment (new page) groups.

Key finding: The new landing page underperformed the control. The treatment group recorded a conversion rate of 11.88% versus 12.04% for control — a relative drop of -1.31%. Statistical testing (two-proportion z-test) returned a p-value of 0.19, which is well above the 0.05 threshold. This means the observed difference is not statistically significant and could be due to random variation. The recommendation is to not deploy the new landing page.

2. Experiment Design

2.1 Objective

To determine whether the new landing page design produces a statistically significant improvement in user conversion rate compared to the existing landing page.

2.2 Hypothesis

Null Hypothesis (H_0): The new landing page has no effect on conversion rate. Any observed difference is due to chance.

Alternative Hypothesis (H_1): The new landing page produces a different conversion rate compared to the control page.

2.3 Experiment Groups

Group	Landing Page	Total Users	Converted Users	Conversion Rate
Control	Old Page	145,274	17,489	12.04%
Treatment	New Page	145,310	17,264	11.88%
Total	—	290,584	34,753	11.96%

3. Dataset Overview

3.1 Raw Data

Attribute	Value
Source File	ab_data.csv
Total Raw Records	294,478 rows
Columns	user_id, timestamp, group, landing_page, converted
Conversion Values	Binary — 0 (not converted) / 1 (converted)
Experiment Period	January 2017 (approx. 4 weeks)

3.2 Data Cleaning Steps

- Identified and removed mismatched rows where group and landing page did not align (e.g., control user shown new page). Total mismatched rows removed: ~3,894
- Identified 1 duplicate user_id (773192) appearing twice in the dataset
- Removed duplicates using drop_duplicates on user_id, keeping the first occurrence
- Final cleaned dataset: 290,584 unique users saved as ab_cleaned.csv
- Cleaned data exported to MySQL database (ab_testing_db) for SQL analysis

4. Statistical Analysis

4.1 Sample Ratio Mismatch (SRM) Check

Before running the main statistical test, a Chi-Square goodness-of-fit test was performed to confirm the user split was balanced.

Metric	Value
Expected Split	50% / 50%
Actual Split (Control)	145,274 (49.99%)
Actual Split (Treatment)	145,310 (50.01%)
Chi-Square Result	No significant SRM detected
Interpretation	Randomization was valid — experiment split is acceptable

4.2 Two-Proportion Z-Test

A two-tailed proportions z-test was used to compare conversion rates between the two groups.

Metric	Value
Control Conversion Rate	12.04% (0.120386)
Treatment Conversion Rate	11.88% (0.118808)
Absolute Difference	-0.16%

Metric	Value
Relative Lift	-1.31%
Z-Statistic	-1.31
P-Value	0.19
Significance Threshold (α)	0.05
Result	NOT Statistically Significant ($p > 0.05$)

Interpretation: Since p-value (0.19) > 0.05, we fail to reject the null hypothesis. The difference in conversion rates is not statistically significant. We cannot conclude that the new landing page performs differently from the old page.

4.3 Confidence Intervals

Group	Conversion Rate	95% Confidence Interval
Control	12.04%	(11.87% , 12.21%)
Treatment	11.88%	(11.71% , 12.05%)

The confidence intervals overlap substantially. This confirms there is no clear evidence that the new landing page performs differently from the old landing page. Overlapping CIs reinforce the z-test conclusion.

4.4 Statistical Power Analysis

Power analysis was conducted to assess whether the experiment had sufficient sensitivity to detect a meaningful improvement.

Parameter	Value
Baseline Conversion Rate	12.04%
Minimum Detectable Effect (MDE)	2% relative improvement (target: 12.28%)
Sample Size (per group)	~145,274
Statistical Power	0.80 (80%)
Interpretation	The test had 80% power to detect a 2% relative lift — borderline adequate

Note: The actual observed effect was only -1.31% (a drop, not an improvement). The experiment may lack sufficient power to detect very small positive effects below 2% relative lift, but that is not the deciding factor here — the treatment actually performed worse.

5. Weekly Trend Analysis

Weekly conversion rates were tracked to detect novelty effects (temporary spikes when users first see a new design) or sustained improvement patterns.

Week	Control CR	Treatment CR	Observation
Week 1	11.86%	11.86%	Nearly identical at start
Week 2	11.88%	11.88%	No difference observed
Week 3	12.07%	12.07%	Both groups rise slightly
Week 4	12.21%	12.21%	Partial week — ~10K users each

- No novelty effect was detected — treatment did not spike early and decline
- Both groups showed near-identical weekly trends throughout the experiment
- Week 4 shows reduced user count (~10K vs ~46K) due to partial data collection, not a system failure — the symmetric drop across both groups confirms this
- Control group maintained consistently slightly higher conversion rates across all weeks

6. Business Impact Assessment

Revenue projections were calculated using an assumed conversion value of \$500 per successful conversion, applied uniformly to both groups. These are directional estimates for business decision-making, not actual financial figures.

6.1 Monthly Impact

Metric	Control	Treatment
Assumed Monthly Users	1,000,000	1,000,000
Projected Monthly Conversions	120,386	118,808
Estimated Monthly Revenue	\$60.19M	\$59.40M
Monthly Revenue Difference	—	-\$789,000 loss if treatment deployed

6.2 Annual Impact

Metric	Value
Estimated Monthly Revenue Loss (if deployed)	\$789,000
Estimated Annual Revenue Loss (if deployed)	\$9.47M

Metric	Value
Recommendation	Do Not Deploy — potential ~\$9.47M annual revenue at risk

Important: These revenue figures are projections based on assumed benchmarks. Actual business impact should be validated with real transaction data. The direction of the impact (negative) is consistent with the statistical results.

7. SQL Analysis Summary

The cleaned dataset was exported to a MySQL database (ab_testing_db) and 15 structured queries were written to validate and extend the Python analysis.

Query #	Purpose	Key Output
Q1-Q3	Data exploration	5 rows preview, total count, unique groups
Q4	Duplicate check in SQL	Confirmed no duplicates post-cleaning
Q5-Q6	Group-level user and conversion counts	145,274 control / 145,310 treatment
Q7	Conversion rate per group	Control: 12.04% Treatment: 11.88%
Q8	Relative conversion lift	-1.31% lift (treatment underperformed)
Q9-Q11	Weekly conversion trends	No significant weekly variation detected
Q12-Q13	Monthly conversion and revenue estimates	Revenue projections at 1M monthly users
Q14	Master KPI summary query	Single query for all main experiment metrics
Q15	Power BI export dataset	Weekly aggregated data for dashboard visuals

8. Assumptions and Justifications

Assumption	Reason / Justification
converted = 1 represents a completed desired user action	Dataset provided only binary values (0/1). Value 1 was interpreted as a successful conversion event such as a purchase, sign-up, or form submission.
Users were randomly and independently assigned to groups	Random assignment eliminates selection bias. Independence satisfies the assumptions required by the two-proportion z-test.
\$500 fixed revenue per conversion was assumed	No actual transaction data was available. A constant was applied uniformly to both groups. Revenue figures are

Assumption	Reason / Justification
	directional estimates only and should not be treated as precise financial data.
All revenue metrics are projected estimates	Revenue is derived from the assumed benchmark. It illustrates potential business impact under stated conditions, not actual financial outcomes.
Experiment environment remained stable throughout	Valid A/B testing assumes no major tracking failures, deployment bugs, or instrumentation errors during the test period.
Week 4 drop (10K vs 46K users) is due to partial data collection	The drop is symmetric across both groups, indicating the experiment ended mid-week rather than a biased or technical event.
External factors did not significantly influence outcomes	Isolates the effect of the landing page change from unrelated variables such as marketing campaigns, seasonal events, or abnormal traffic spikes.
Significance threshold (alpha) = 0.05 was used	The 5% significance level is the widely accepted industry standard in A/B experimentation. $p > 0.05$ means we fail to reject the null hypothesis.
Sample ratio was acceptable (49.99% vs 50.01%)	Near-perfect 50/50 split confirmed via Chi-Square test. No sample ratio mismatch detected, preserving the validity of the comparison.
Weekly aggregation was sufficient for trend analysis	With ~42K-46K users per group per week, weekly granularity provides statistically stable data points without the noise of daily fluctuations.
Experiment conclusions are based on conversion rate and statistical significance only	Scope is analytical. Broader factors like implementation cost, UX quality, and long-term retention were outside the dataset and are not factored into the verdict.

9. Conclusion and Recommendation

The A/B test was conducted rigorously with a near-perfect 50/50 user split across 290,584 users over 4 weeks. Despite a large sample size, the new landing page failed to demonstrate any meaningful improvement in conversion rate.

Decision Factor	Result	Implication
Conversion Rate Lift	-1.31%	Treatment performed worse
P-Value	0.19 (> 0.05)	Not statistically significant
Z-Statistic	-1.31	Negative, confirms underperformance
Confidence Intervals	Overlapping	No meaningful difference confirmed
Statistical Power	0.80	Adequate for 2% MDE — not the limiting factor
Weekly Trend	No novelty effect	No sustained improvement at any point

Decision Factor	Result	Implication
Revenue Impact	-\$9.47M/year	Deploying = potential revenue loss

Final Recommendation: Do Not Ship.

The new landing page should not be deployed. The evidence — a negative lift, non-significant p-value, overlapping confidence intervals, and consistent underperformance across all 4 weeks — collectively indicates that the treatment page offers no benefit. Deploying it carries a projected annual revenue risk of approximately \$9.47M. The product or design team should revisit the landing page concept with user research and test a fundamentally different variant before running another experiment.

10. Tools and Workflow

Tool	Purpose
Python (Pandas, NumPy)	Data loading, cleaning, EDA, and statistical testing
SciPy / Statsmodels	Chi-Square test, Z-test, Confidence Intervals, Power Analysis
Matplotlib	Weekly conversion trend visualization
MySQL (SQLAlchemy + PyMySQL)	Data export, SQL-based analysis, KPI queries
Power BI	Interactive dashboard — KPI cards, trend charts, user distribution, revenue metrics

**Revenue metrics are projected estimates based on assumed conversion value benchmarks (\$500 per conversion). Actual revenue impact requires validation with real transaction data.*